

ITS203 Object-Oriented Design and Programming

Assessment C – Milestone 1: Project Idea Proposal

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Project Title: StudyTrack: Student Assignment & Study Planner

1. Problem Description

Students usually have homework assignments, study projects due, and many different topics to keep track of. Having this info written down anywhere (notebooks, calendars, messages, etc.) or different applications can get hectic trying to keep track of what is due when. We may forget deadlines, become messy in organizations, and not know how to prioritize work. This application is intended for students who need an easy method to organize all their academic assignments/study tasks. Students will be able to add assignments/tasks to list what subject it's for, title of assignment/task, due date, priority, and if it's completed. User-friendly graphics will display this information for users to see what they need to do.

2. Proposed Idea

The project idea is going to be a desktop application that utilizes Windows Forms. The name of our project will be StudyTrack: Student Assignment & Study Planner. This application will allow students to have a centralized location to create, view, update, and manage school related tasks. Every task will contain a title, category, due date, priority, and completion status. Students will be able to create new tasks, edit past tasks, complete tasks, and delete tasks. The application will allow for basic organizational functions. Students will be able to view tasks by priority or due date. Validation will be used to prevent improper user input like entering a blank title or invalid date. The application will be written in the C# language and use Windows Forms. We will use an object-oriented design. Classes will be used to represent the major objects and responsibilities within the application. The base features that we will focus on will be just the management of tasks.

We will not try to build an entire learning management system. This will allow us to finish within our assigned timeframe.

3. Inspiration

Study Track is based off need I found myself having this trimester. Due dates and information about assignments are spread throughout Moodle, unit outlines, and emails. I have found myself making temporary lists in notepads apps to remember what is due when. Throughout this, I realized that there should be a lightweight planner specifically for schoolwork. StudyTrack will be different from your typical to-do app. It will allow me to organize subjects and task types (e.g. homework, quizzes, tests, reading). This allows for better organization of information we as students need to constantly be aware of. I chose this project because it solves a problem I have every week. I will have a strong drive to continue working on and using this app after this unit is over.

4. Functional Requirements

The application will provide the following core functionality:

1. **Add a task:** The user can add a new assignment or study item by typing in the title of the task, topic, due date, and importance.
2. **View tasks:** Through the Windows Forms interface, the user can see all tasks that are stored at the moment and important information about each one.
3. **Edit a task:** User can choose a current task and update information about it if needed. This would apply if there were changes to the requirements or due dates.
4. **Delete a task:** User can delete unnecessary tasks.
5. **Mark a task as completed:** Task completion can be toggled by the user once the work is completed.
6. **Set task priority:** The user can specify a priority for a task of Low, Medium, or High.
7. **Organize tasks by due date or priority:** Tasks can be sorted in a way that makes it easy for the user to see what tasks need attention.
8. **Validate user input:** The app will validate key fields and show a useful message when invalid/incomplete data is entered.

5. Non-functional Requirements

The application should meet the following non-functional requirements:

- **Usability:** Interfaces should be intuitive enough that a first-time user needs little explanation.
- **Performance:** Basic operations like adding, editing, deleting and viewing tasks should be instant.

- **Reliability:** Invalid user input should never break the application, and the application should give helpful feedback in these cases.
- **Maintenance:** The code should be logically separated with clear class structures and responsibilities. This will allow the application to be changed or added to with less difficulty.
- **Data persistence:** Tasks should persist between closing and reopening the application, if they have been saved properly.

6. Data Storage Plan

The application will be using file-based storage instead of a database. The application is designed to handle a user's personal collection of subjects and tasks, which will not reach very high numbers. JSON format was chosen because it allows the storage mechanism to be simple, human readable, and easily backed up. The data will be held locally in two JSON files located in the application's data folder. These files will be loaded into memory on startup and re-written to the documents folder any time the user adds, edits, deletes, or completes an item.

The **subjects.json** file will store a list of subject objects. Each subject will contain:

- SubjectId - integer to uniquely identify the subject.
- Name - string containing name of the subject.
- Code - string containing code of the subject.
- ColorTag - string containing a hexadecimal color code which will be used to color tag the subject.

The **tasks.json** file will store a list of task objects. Each task will contain:

- TaskId – an integer value that uniquely identifies the task.
- Title – string value with the name of the task.
- SubjectId – integer value that connects the task to a subject.
- DueDate – date value when the task is due.
- Priority – string value with options of Low, Medium or High.
- TaskType – string value with options for Assignment, Quiz, Exam, Reading or Other.
- IsComplete – Boolean value if the task has been completed or not.
- Notes – optional string value with notes regarding the task.

Tasks will reference subjects by SubjectId instead of storing the subjects name and colour in every task. This ensures that the information will remain synchronized if we ever need to

edit a subject. It also allows us to implement the future object-oriented design of the application. We will have a Subject class and a Task class. Every Task will have a Subject.